Student:	Email:
Isaiah Mosley	isaiahmosley80@gmail.com

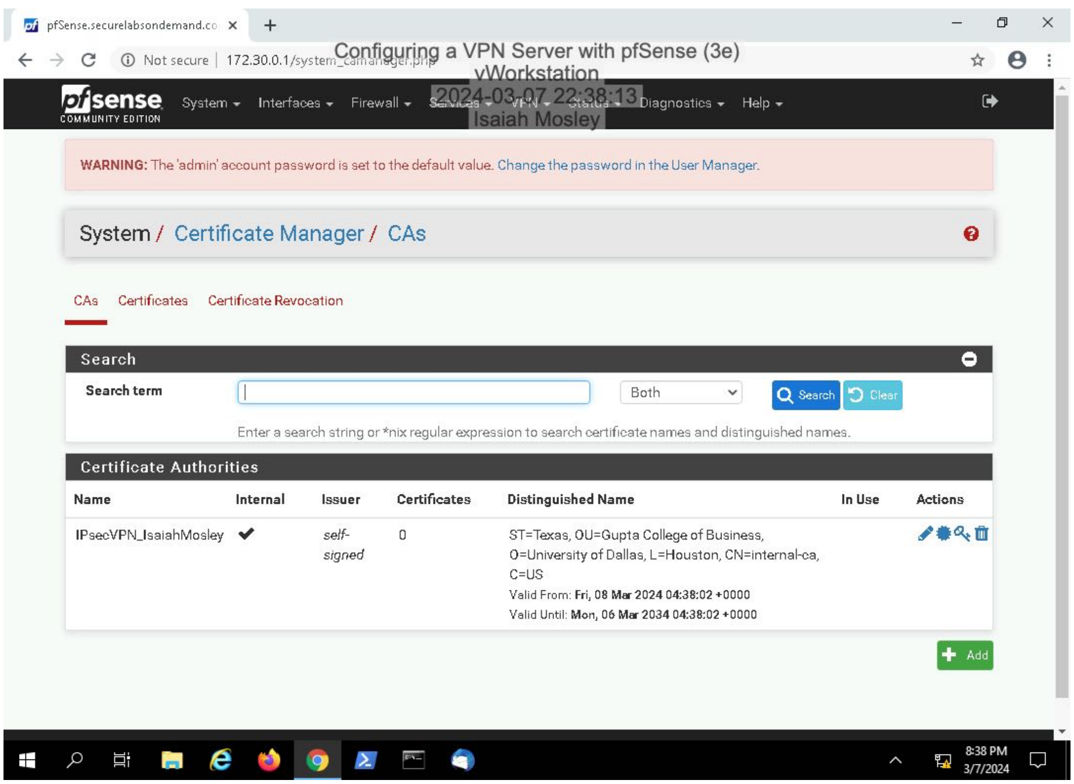
Time on Task:	Progress:
6 hours, 30 minutes	100%

Report Generated: Sunday, March 10, 2024 at 1:04 AM

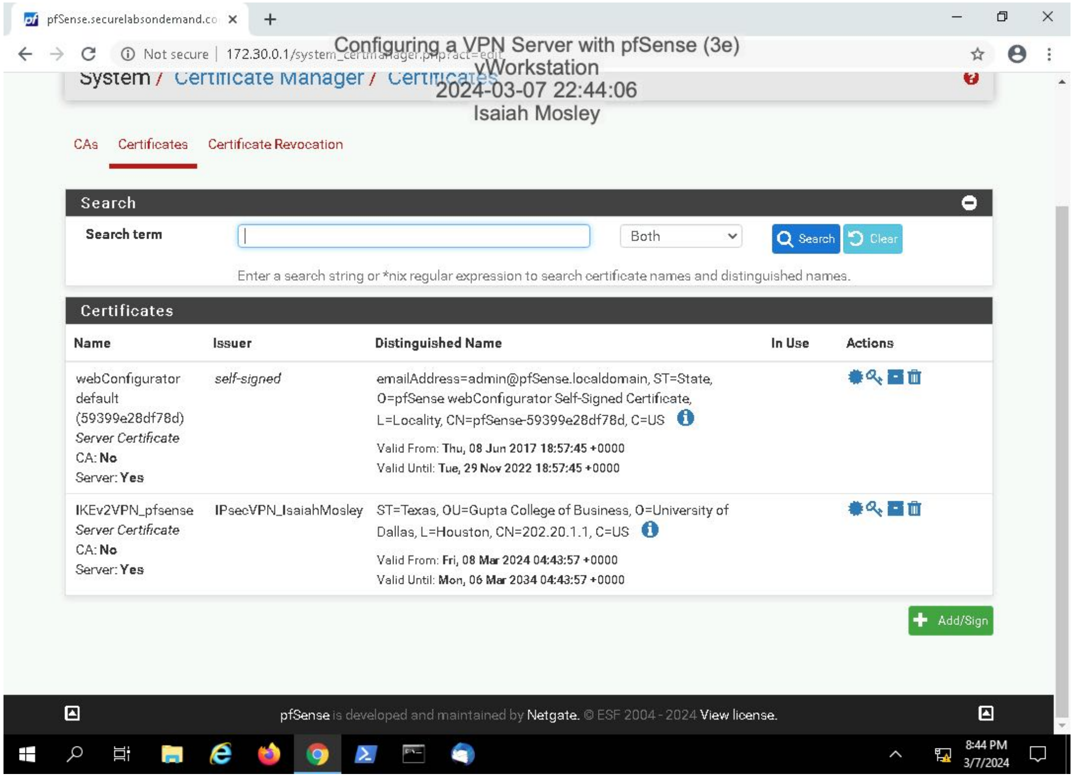
Section 1: Hands-On Demonstration

Part 1: Configure an IPsec VPN Server

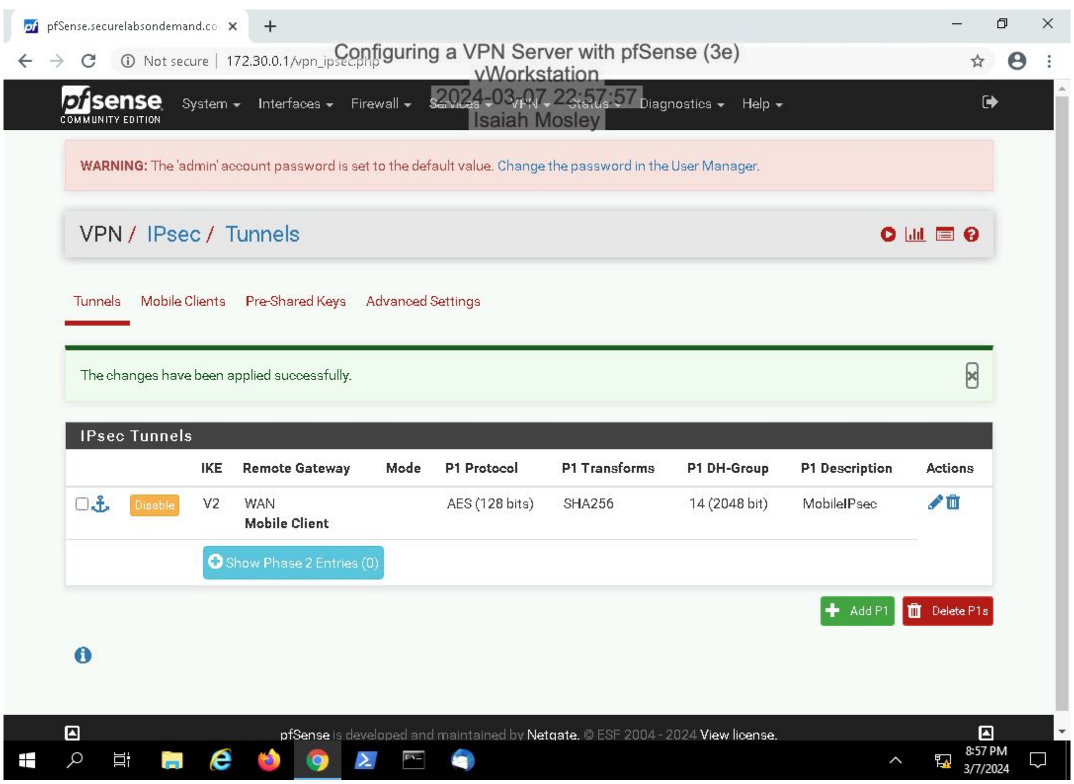
18. **Make a screen capture** showing the **updated Certificate Authorities table**.



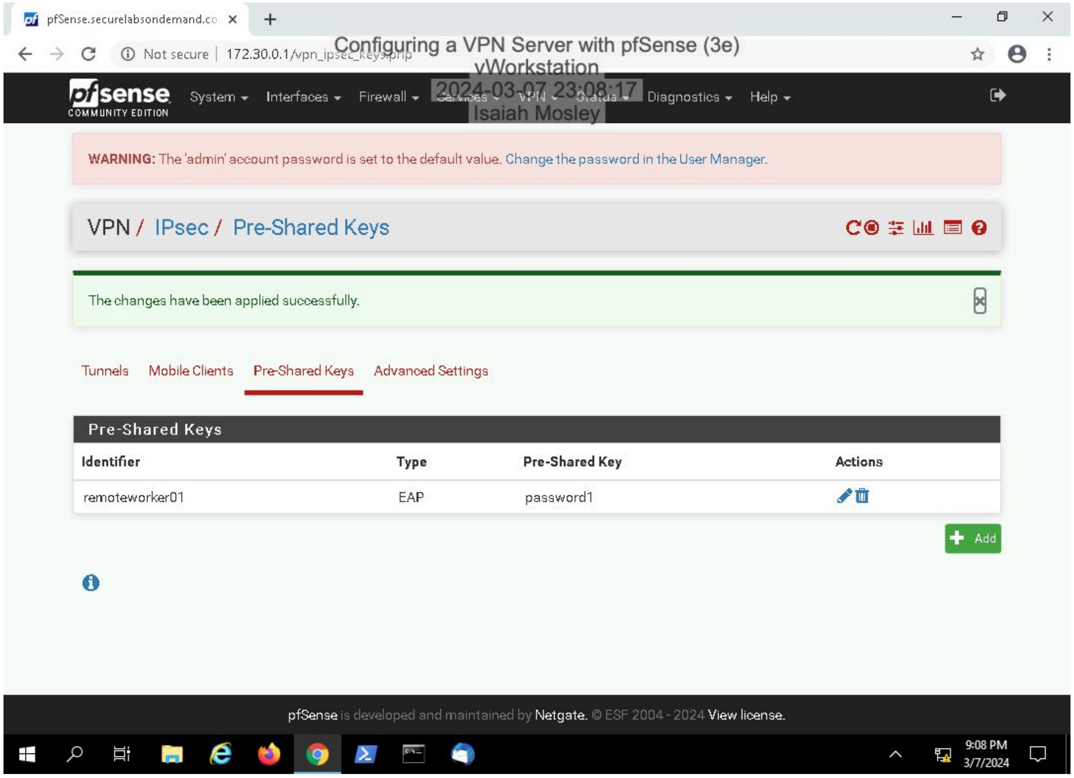
32. **Make a screen capture** showing the **updated Certificates table**.



56. **Make a screen capture** showing the **updated IPsec Tunnels table.**

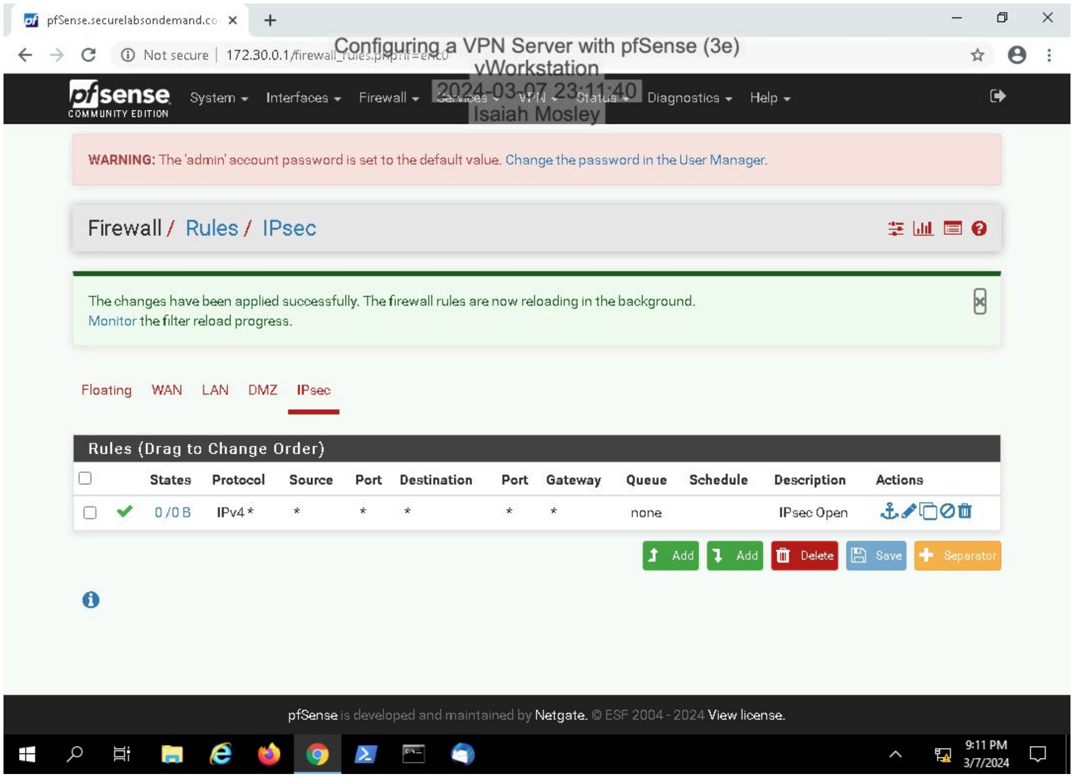


74. **Make a screen capture** showing the **updated Pre-Shared Keys table**.



Part 2: Configure a Firewall Rule for VPN Traffic

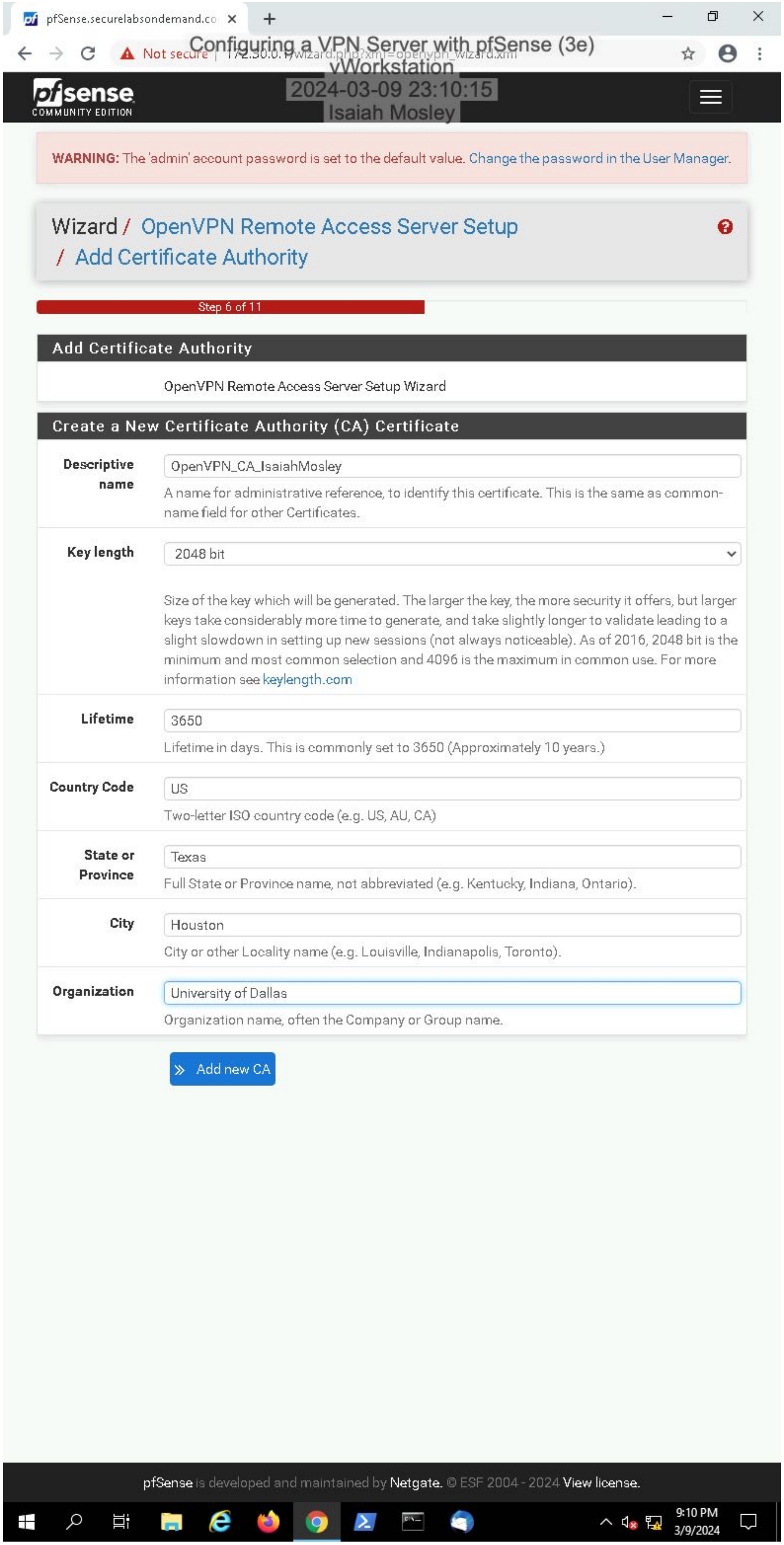
9. **Make a screen capture** showing the **updated IPsec Rules table**.



Section 2: Applied Learning

Part 1: Configure an OpenVPN VPN Server

13. **Make a screen capture** showing the **CA configuration form.**


pfSense.securelabsondemand.co
Not secure | 172.30.0.1/wizard.php?xml=openvpn_wizard.xml
Configuring a VPN Server with pfSense (3e)
vWorkstation
2024-03-09 23:10:15
Isaiah Mosley
pfsense
COMMUNITY EDITION
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Wizard / OpenVPN Remote Access Server Setup / Add Certificate Authority
Step 6 of 11
Add Certificate Authority
OpenVPN Remote Access Server Setup Wizard
Create a New Certificate Authority (CA) Certificate
Descriptive name
OpenVPN_CA_IsaiahMosley
A name for administrative reference, to identify this certificate. This is the same as common-name field for other Certificates.
Key length
2048 bit
Size of the key which will be generated. The larger the key, the more security it offers, but larger keys take considerably more time to generate, and take slightly longer to validate leading to a slight slowdown in setting up new sessions (not always noticeable). As of 2016, 2048 bit is the minimum and most common selection and 4096 is the maximum in common use. For more information see keylength.com
Lifetime
3650
Lifetime in days. This is commonly set to 3650 (Approximately 10 years.)
Country Code
US
Two-letter ISO country code (e.g. US, AU, CA)
State or Province
Texas
Full State or Province name, not abbreviated (e.g. Kentucky, Indiana, Ontario).
City
Houston
City or other Locality name (e.g. Louisville, Indianapolis, Toronto).
Organization
University of Dallas
Organization name, often the Company or Group name.
Add new CA
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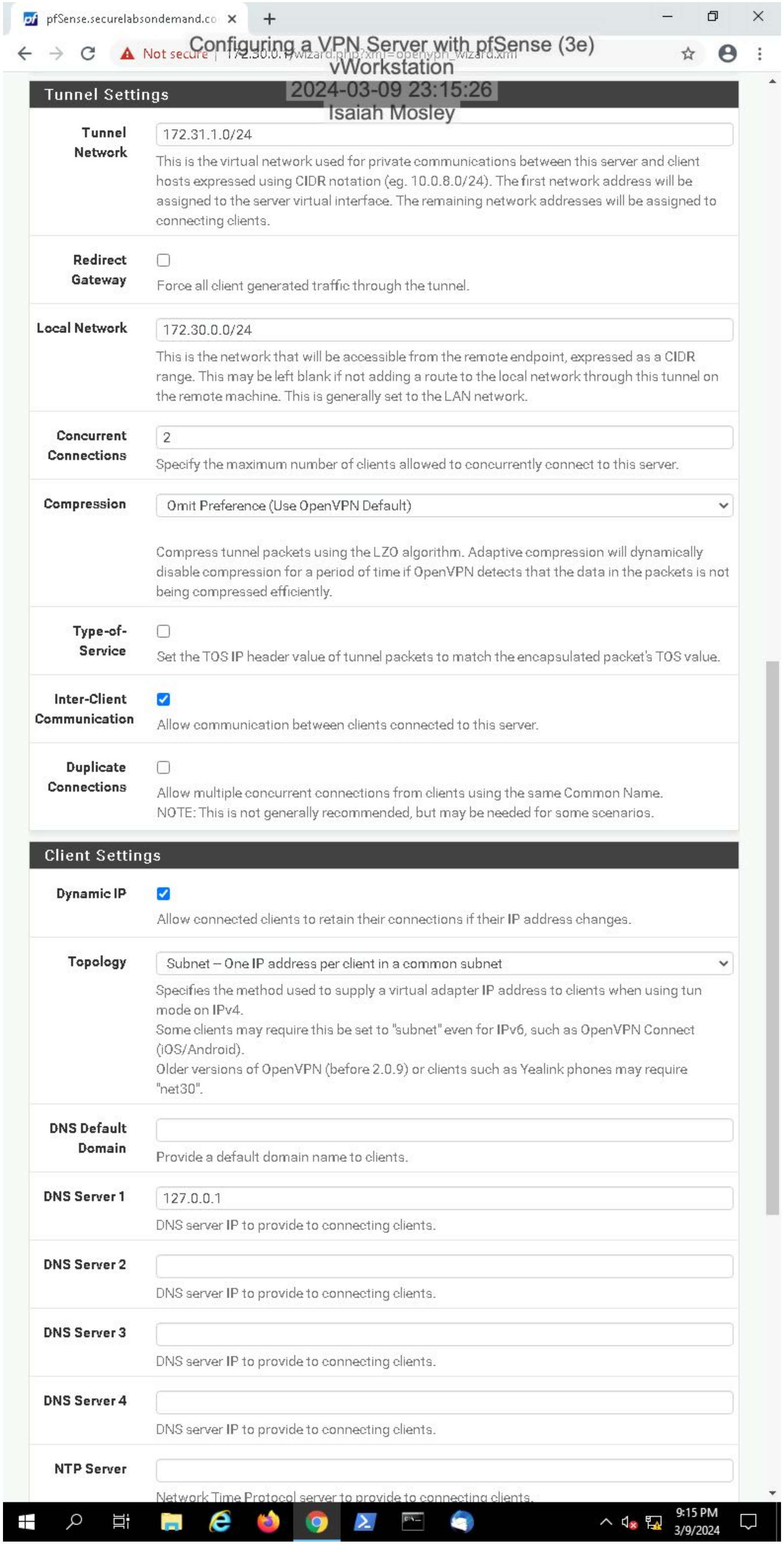
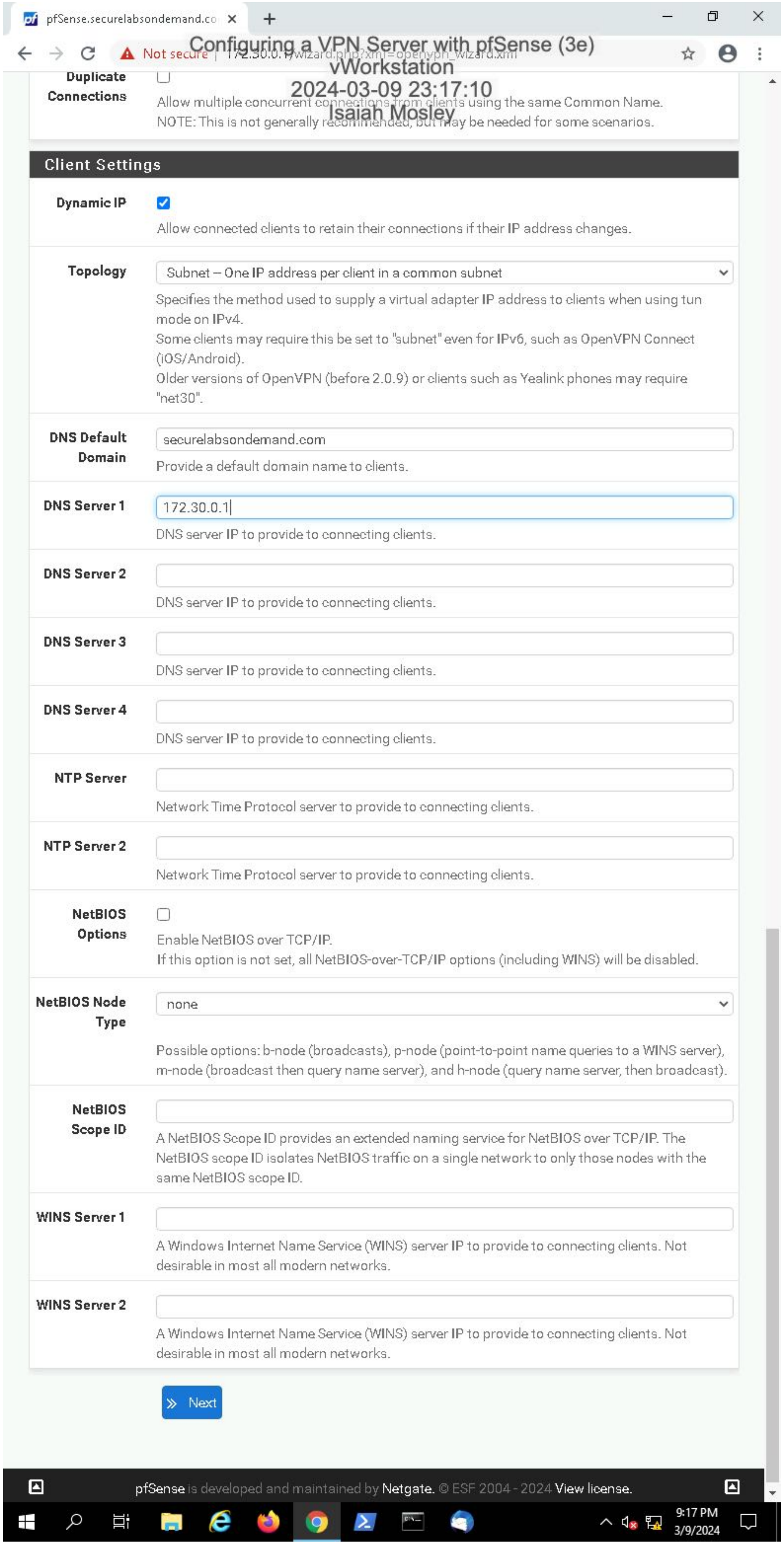
9:10 PM
3/9/2024

27. **Make a screen capture** showing the **Tunnel Settings section**.


pfSense.securelabsondemand.co
Not secure | 172.30.0.1/wizard.php?xml=openvpn_wizard.xml
Configuring a VPN Server with pfSense (3e)
vWorkstation
2024-03-09 23:15:26
Isaiah Mosley
Tunnel Settings
Tunnel Network
172.31.1.0/24
This is the virtual network used for private communications between this server and client hosts expressed using CIDR notation (eg. 10.0.8.0/24). The first network address will be assigned to the server virtual interface. The remaining network addresses will be assigned to connecting clients.
Redirect Gateway
Force all client generated traffic through the tunnel.
Local Network
172.30.0.0/24
This is the network that will be accessible from the remote endpoint, expressed as a CIDR range. This may be left blank if not adding a route to the local network through this tunnel on the remote machine. This is generally set to the LAN network.
Concurrent Connections
2
Specify the maximum number of clients allowed to concurrently connect to this server.
Compression
Omit Preference (Use OpenVPN Default)
Compress tunnel packets using the LZO algorithm. Adaptive compression will dynamically disable compression for a period of time if OpenVPN detects that the data in the packets is not being compressed efficiently.
Type-of-Service
Set the TOS IP header value of tunnel packets to match the encapsulated packet's TOS value.
Inter-Client Communication
Allow communication between clients connected to this server.
Duplicate Connections
Allow multiple concurrent connections from clients using the same Common Name.
NOTE: This is not generally recommended, but may be needed for some scenarios.
Client Settings
Dynamic IP
Allow connected clients to retain their connections if their IP address changes.
Topology
Subnet – One IP address per client in a common subnet
Specifies the method used to supply a virtual adapter IP address to clients when using tun mode on IPv4.
Some clients may require this be set to "subnet" even for IPv6, such as OpenVPN Connect (iOS/Android).
Older versions of OpenVPN (before 2.0.9) or clients such as Yealink phones may require "net30".
DNS Default Domain
Provide a default domain name to clients.
DNS Server 1
127.0.0.1
DNS server IP to provide to connecting clients.
DNS Server 2
DNS server IP to provide to connecting clients.
DNS Server 3
DNS server IP to provide to connecting clients.
DNS Server 4
DNS server IP to provide to connecting clients.
NTP Server
Network Time Protocol server to provide to connecting clients.
9:15 PM
3/9/2024

30. **Make a screen capture** showing the **Client Settings section.**



pfSense.securelabsondemand.co
Not secure | 172.30.0.1/wizard.php?xml=openvpn_wizard.xml
Configuring a VPN Server with pfSense (3e)
vWorkstation
2024-03-09 23:17:10
Isaiah Mosley
Duplicate Connections
Allow multiple concurrent connections from clients using the same Common Name.
NOTE: This is not generally recommended, but may be needed for some scenarios.
Client Settings
Dynamic IP
Allow connected clients to retain their connections if their IP address changes.
Topology
Subnet – One IP address per client in a common subnet
Specifies the method used to supply a virtual adapter IP address to clients when using tun mode on IPv4.
Some clients may require this be set to "subnet" even for IPv6, such as OpenVPN Connect (iOS/Android).
Older versions of OpenVPN (before 2.0.9) or clients such as Yealink phones may require "net30".
DNS Default Domain
securelabsondemand.com
Provide a default domain name to clients.
DNS Server 1
172.30.0.1
DNS server IP to provide to connecting clients.
DNS Server 2
DNS server IP to provide to connecting clients.
DNS Server 3
DNS server IP to provide to connecting clients.
DNS Server 4
DNS server IP to provide to connecting clients.
NTP Server
Network Time Protocol server to provide to connecting clients.
NTP Server 2
Network Time Protocol server to provide to connecting clients.
NetBIOS Options
Enable NetBIOS over TCP/IP.
If this option is not set, all NetBIOS-over-TCP/IP options (including WINS) will be disabled.
NetBIOS Node Type
none
Possible options: b-node (broadcasts), p-node (point-to-point name queries to a WINS server), m-node (broadcast then query name server), and h-node (query name server, then broadcast).
NetBIOS Scope ID
A NetBIOS Scope ID provides an extended naming service for NetBIOS over TCP/IP. The NetBIOS scope ID isolates NetBIOS traffic on a single network to only those nodes with the same NetBIOS scope ID.
WINS Server 1
A Windows Internet Name Service (WINS) server IP to provide to connecting clients. Not desirable in most all modern networks.
WINS Server 2
A Windows Internet Name Service (WINS) server IP to provide to connecting clients. Not desirable in most all modern networks.
Next
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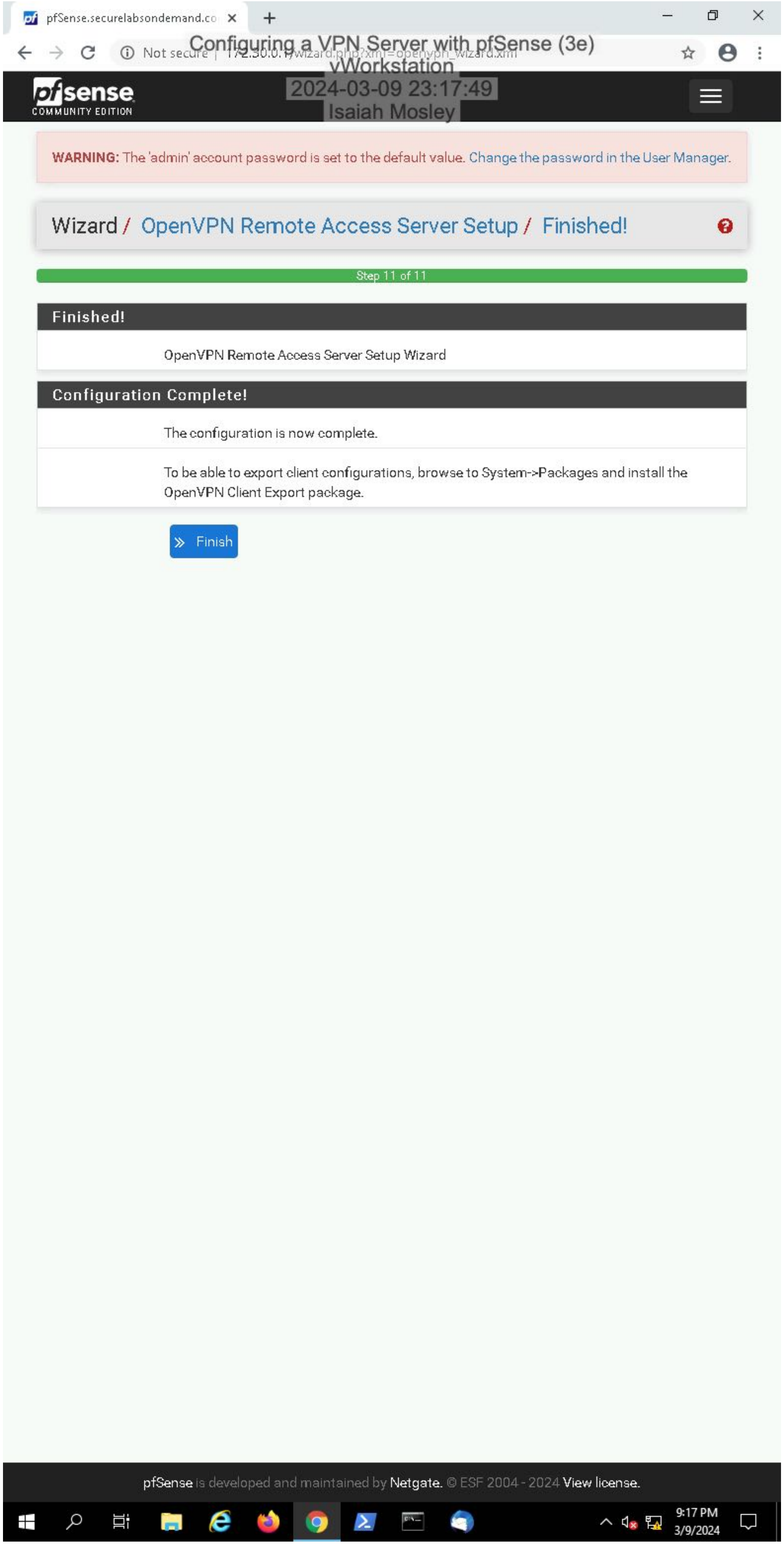
9:17 PM
3/9/2024

Part 2: Configure a Firewall Rule for VPN Traffic

2. **Make a screen capture** showing the **completed OpenVPN configuration**.


pfSense.securelabsondemand.co
Not secure | 172.30.0.1/wizard.php?xml=openvpn_wizard.xml
Configuring a VPN Server with pfSense (3e)
vWorkstation
2024-03-09 23:17:49
Isaiah Mosley
pfsense COMMUNITY EDITION
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Wizard / OpenVPN Remote Access Server Setup / Finished!
Step 11 of 11
Finished!
OpenVPN Remote Access Server Setup Wizard
Configuration Complete!
The configuration is now complete.
To be able to export client configurations, browse to System->Packages and install the OpenVPN Client Export package.
Finish
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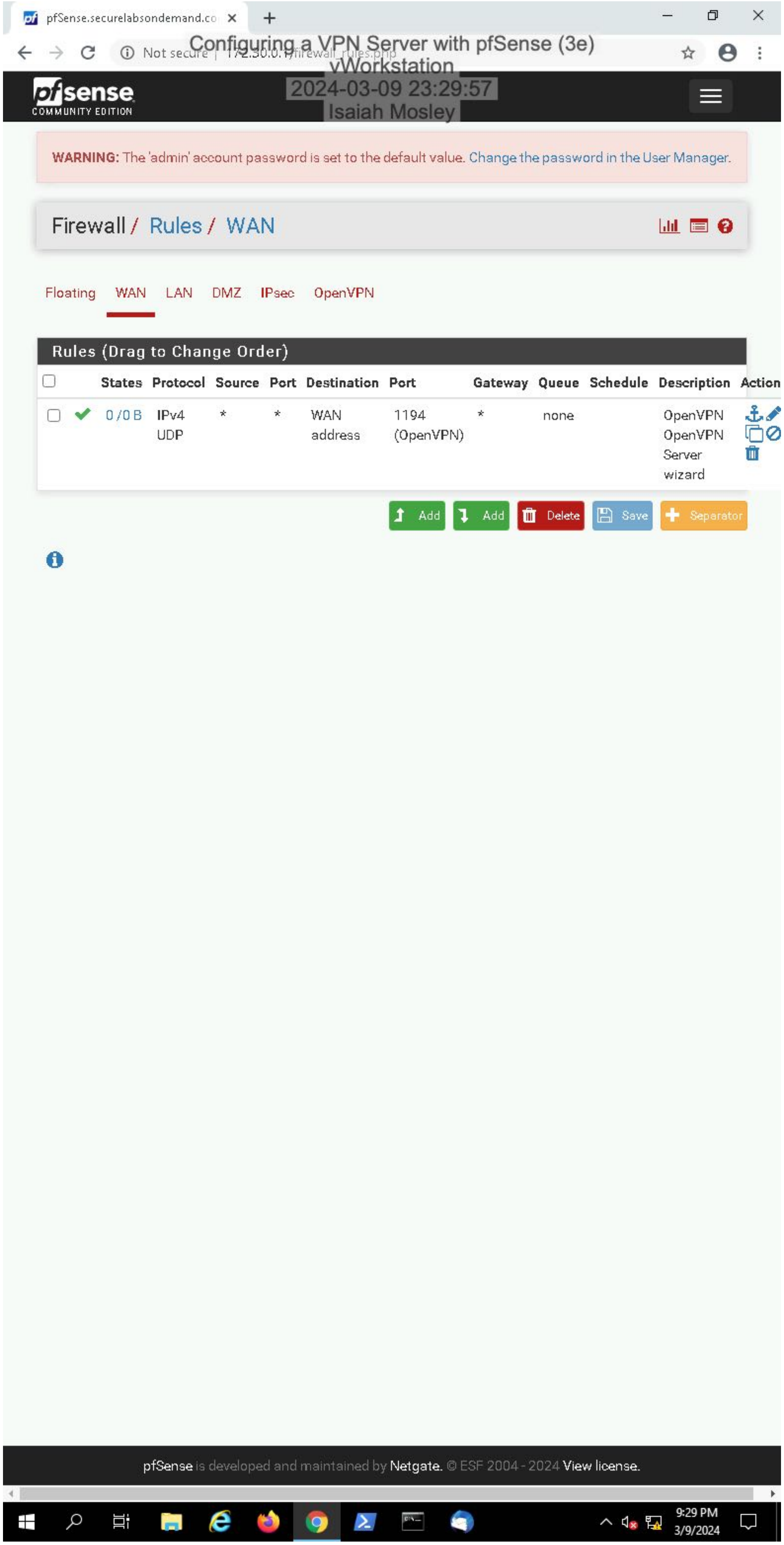
9:17 PM
3/9/2024

5. **Make a screen capture** showing the **OpenVPN rule on the WAN Rules table.**


Configuring a VPN Server with pfSense (3e)
vWorkstation
2024-03-09 23:29:57
Isaiah Mosley
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Firewall / Rules / WAN
Floating
WAN
LAN
DMZ
IPsec
OpenVPN
Rules (Drag to Change Order)
States
Protocol
Source
Port
Destination
Port
Gateway
Queue
Schedule
Description
Action
0 /0 B
IPv4 UDP
*
*
WAN address
1194 (OpenVPN)
*
none
OpenVPN OpenVPN Server wizard
Add
Add
Delete
Save
Separator
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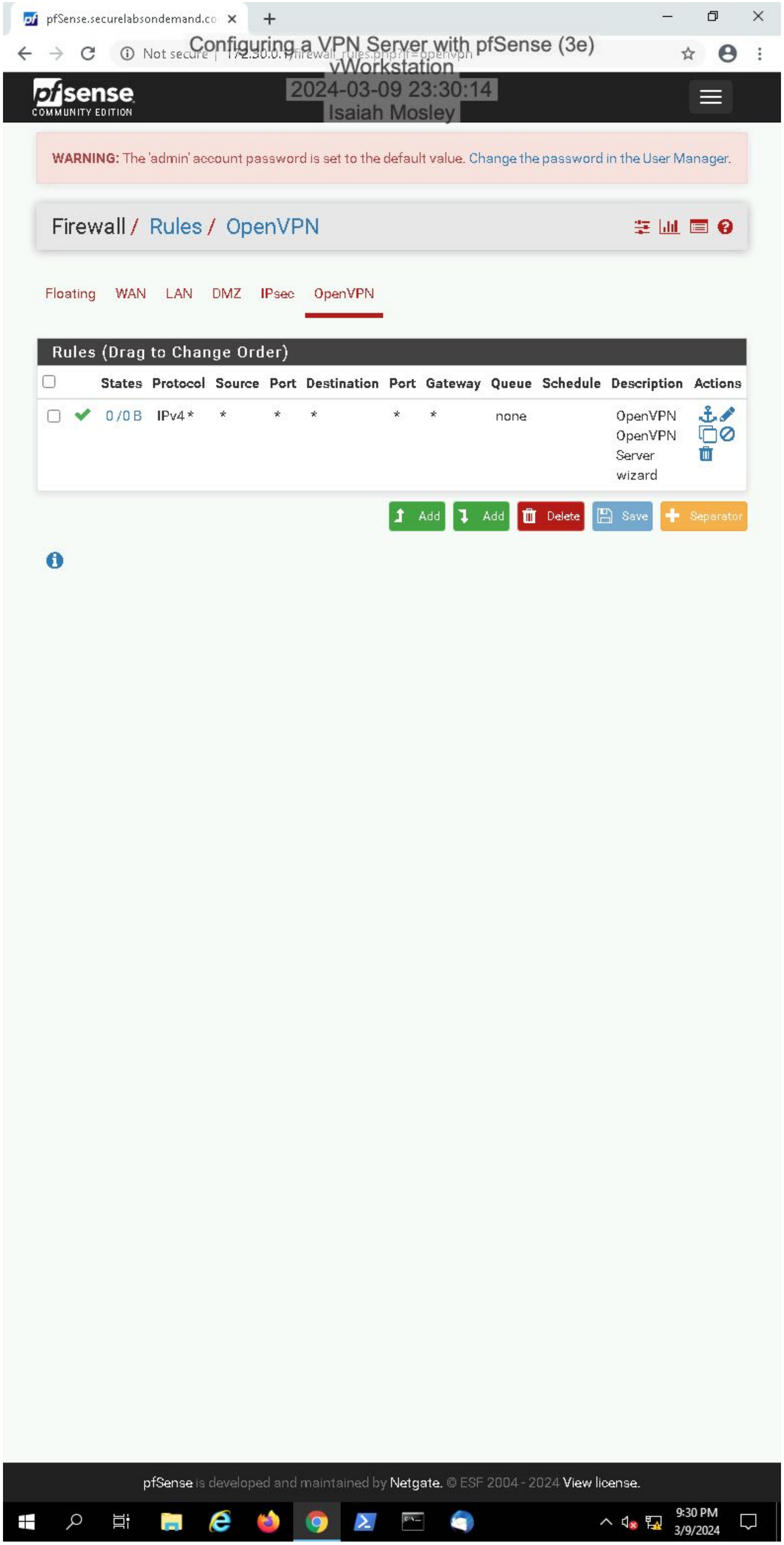
9:29 PM
3/9/2024

7. **Make a screen capture** showing the **OpenVPN rule on the OpenVPN Rules table**.


pfSense.securelabsondemand.co
Not secure | 172.30.0.1/firewall_rules.php?if=openvpn
Configuring a VPN Server with pfSense (3e)
vWorkstation
2024-03-09 23:30:14
Isaiah Mosley
pfsense
COMMUNITY EDITION
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Firewall / Rules / OpenVPN
Floating
WAN
LAN
DMZ
IPsec
OpenVPN
Rules (Drag to Change Order)
States
Protocol
Source
Port
Destination
Port
Gateway
Queue
Schedule
Description
Actions
0 /0 B
IPv4 *
*
*
*
*
*
none
OpenVPN OpenVPN Server wizard
Add
Add
Delete
Save
Separator
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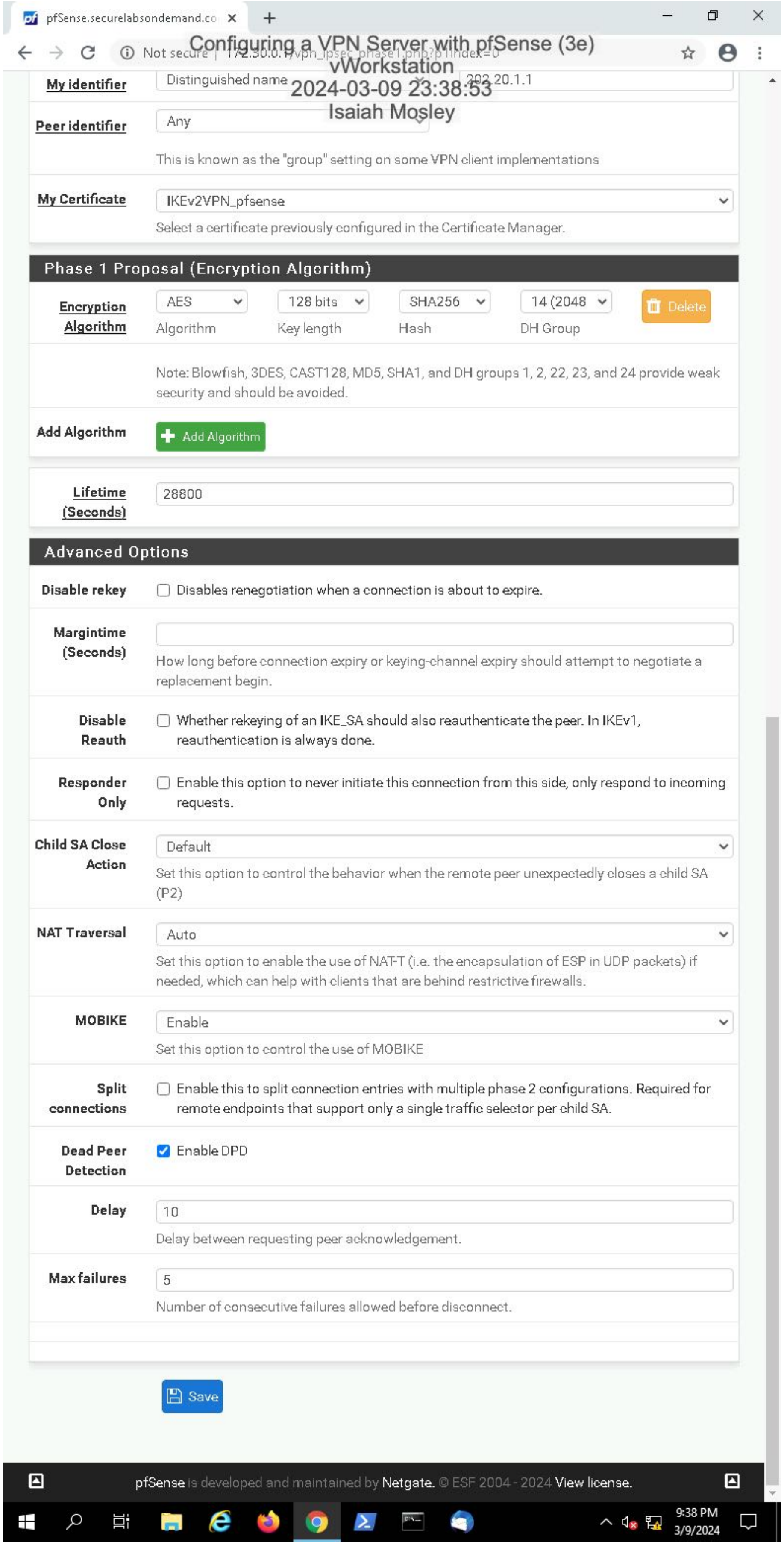
9:30 PM
3/9/2024

Section 3: Challenge and Analysis

Part 1: Enable IP Roaming for Remote VPN Clients

Make a screen capture showing the **enabled MOBIKE option in the IPsec tunnel configuration**.



Configuring a VPN Server with pfSense (3e)
vWorkstation
2024-03-09 23:38:53
Isaiah Mosley
My identifier
Distinguished name
Peer identifier
Any
This is known as the "group" setting on some VPN client implementations
My Certificate
IKEv2VPN_pfsense
Select a certificate previously configured in the Certificate Manager.
Phase 1 Proposal (Encryption Algorithm)
Encryption Algorithm
AES
Algorithm
128 bits
Key length
SHA256
Hash
14 (2048
DH Group
Delete
Note: Blowfish, 3DES, CAST128, MD5, SHA1, and DH groups 1, 2, 22, 23, and 24 provide weak security and should be avoided.
Add Algorithm
Add Algorithm
Lifetime (Seconds)
28800
Advanced Options
Disable rekey
Disables renegotiation when a connection is about to expire.
Margintime (Seconds)
How long before connection expiry or keying-channel expiry should attempt to negotiate a replacement begin.
Disable Reauth
Whether rekeying of an IKE_SA should also reauthenticate the peer. In IKEv1, reauthentication is always done.
Responder Only
Enable this option to never initiate this connection from this side, only respond to incoming requests.
Child SA Close Action
Default
Set this option to control the behavior when the remote peer unexpectedly closes a child SA (P2)
NAT Traversal
Auto
Set this option to enable the use of NAT-T (i.e. the encapsulation of ESP in UDP packets) if needed, which can help with clients that are behind restrictive firewalls.
MOBIKE
Enable
Set this option to control the use of MOBIKE
Split connections
Enable this to split connection entries with multiple phase 2 configurations. Required for remote endpoints that support only a single traffic selector per child SA.
Dead Peer Detection
Enable DPD
Delay
10
Delay between requesting peer acknowledgement.
Max failures
5
Number of consecutive failures allowed before disconnect.
Save
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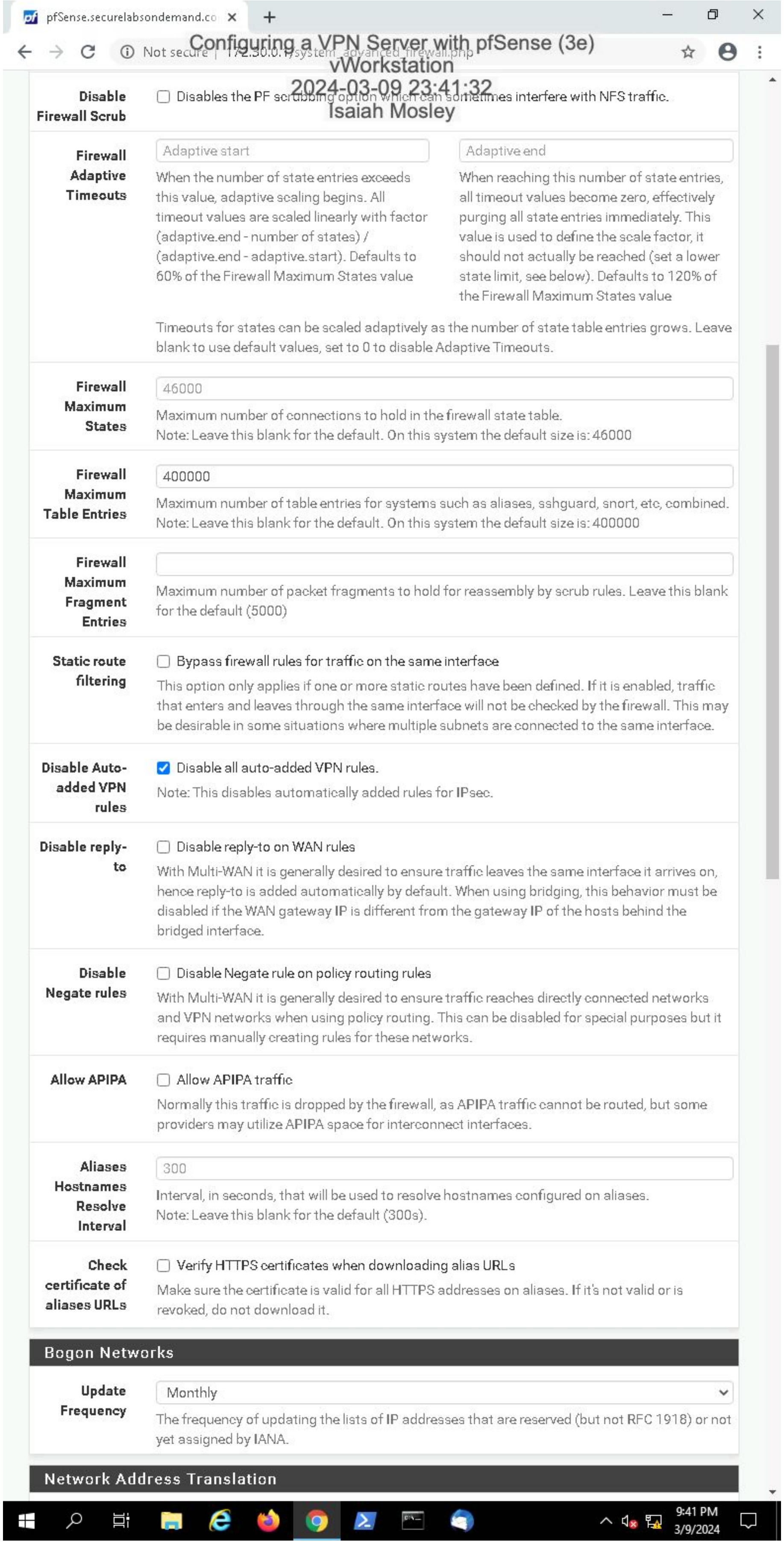
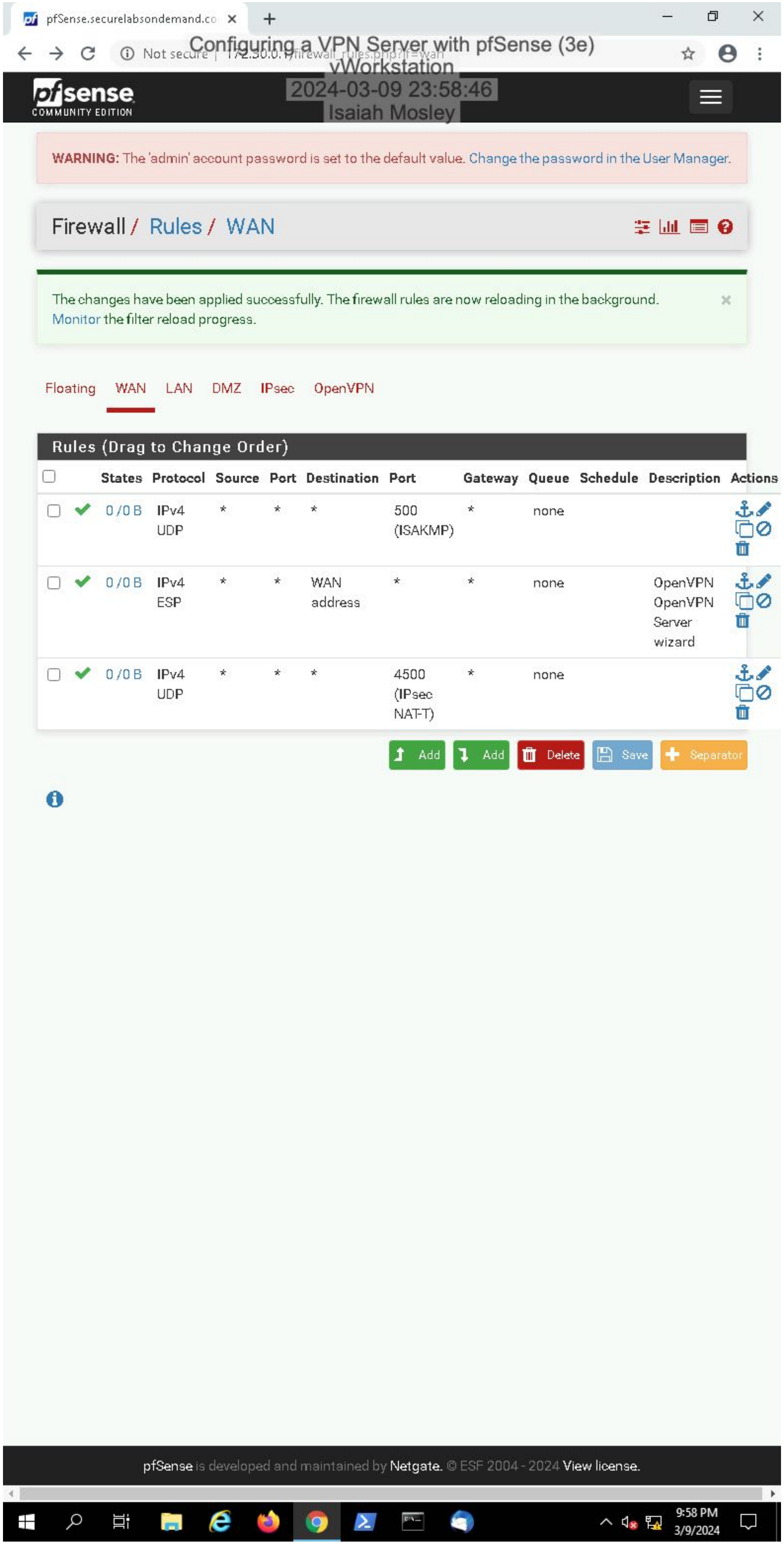
9:38 PM
3/9/2024

Part 2: Create Explicit Firewall Rules for an IPsec VPN

Make a screen capture showing the **disabled automatic IPsec rule creation option**.


pfSense.securelabsondemand.co
Not secure | 172.30.0.1/system_advanced_firewall.php
Configuring a VPN Server with pfSense (3e)
vWorkstation
2024-03-09 23:41:32
Isaiah Mosley
Disable Firewall Scrub
Disables the PF scrubbing option which can sometimes interfere with NFS traffic.
Firewall Adaptive Timeouts
Adaptive start
Adaptive end
When the number of state entries exceeds this value, adaptive scaling begins. All timeout values are scaled linearly with factor (adaptive.end - number of states) / (adaptive.end - adaptive.start). Defaults to 60% of the Firewall Maximum States value
When reaching this number of state entries, all timeout values become zero, effectively purging all state entries immediately. This value is used to define the scale factor, it should not actually be reached (set a lower state limit, see below). Defaults to 120% of the Firewall Maximum States value
Timeouts for states can be scaled adaptively as the number of state table entries grows. Leave blank to use default values, set to 0 to disable Adaptive Timeouts.
Firewall Maximum States
46000
Maximum number of connections to hold in the firewall state table.
Note: Leave this blank for the default. On this system the default size is: 46000
Firewall Maximum Table Entries
400000
Maximum number of table entries for systems such as aliases, sshguard, snort, etc, combined.
Note: Leave this blank for the default. On this system the default size is: 400000
Firewall Maximum Fragment Entries
Maximum number of packet fragments to hold for reassembly by scrub rules. Leave this blank for the default (5000)
Static route filtering
Bypass firewall rules for traffic on the same interface
This option only applies if one or more static routes have been defined. If it is enabled, traffic that enters and leaves through the same interface will not be checked by the firewall. This may be desirable in some situations where multiple subnets are connected to the same interface.
Disable Auto-added VPN rules
Disable all auto-added VPN rules.
Note: This disables automatically added rules for IPsec.
Disable reply-to
Disable reply-to on WAN rules
With Multi-WAN it is generally desired to ensure traffic leaves the same interface it arrives on, hence reply-to is added automatically by default. When using bridging, this behavior must be disabled if the WAN gateway IP is different from the gateway IP of the hosts behind the bridged interface.
Disable Negate rules
Disable Negate rule on policy routing rules
With Multi-WAN it is generally desired to ensure traffic reaches directly connected networks and VPN networks when using policy routing. This can be disabled for special purposes but it requires manually creating rules for these networks.
Allow APIPA
Allow APIPA traffic
Normally this traffic is dropped by the firewall, as APIPA traffic cannot be routed, but some providers may utilize APIPA space for interconnect interfaces.
Aliases Hostnames Resolve Interval
300
Interval, in seconds, that will be used to resolve hostnames configured on aliases.
Note: Leave this blank for the default (300s).
Check certificate of aliases URLs
Verify HTTPS certificates when downloading alias URLs
Make sure the certificate is valid for all HTTPS addresses on aliases. If it's not valid or is revoked, do not download it.
Bogon Networks
Update Frequency
Monthly
The frequency of updating the lists of IP addresses that are reserved (but not RFC 1918) or not yet assigned by IANA.
Network Address Translation
9:41 PM
3/9/2024

Make a screen capture showing your **firewall rules that permit IPsec traffic**.



Configuring a VPN Server with pfSense (3e)
vWorkstation
2024-03-09 23:58:46
Isaiah Mosley
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Firewall / Rules / WAN
The changes have been applied successfully. The firewall rules are now reloading in the background. Monitor the filter reload progress.
Floating WAN LAN DMZ IPsec OpenVPN
Rules (Drag to Change Order)
States Protocol Source Port Destination Port Gateway Queue Schedule Description Actions
0 /0 B IPv4 UDP * * * 500 (ISAKMP) * none
0 /0 B IPv4 ESP * * WAN address * * none OpenVPN OpenVPN Server wizard
0 /0 B IPv4 UDP * * * 4500 (IPsec NAT-T) * none
Add Add Delete Save Separator
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9:58 PM
3/9/2024